02 — Scientific and Technical Plan

TriadicFrameworks: An Open-Science Substrate for Regime-Aware Modeling, Scanning, and Simulation

Submitted to NASA High Priority Open-Source Science (HPOSS)

---

1. Introduction and Motivation

Modern scientific workflows increasingly rely on heterogeneous sensing systems, simulation environments, and data-driven models. These systems often lack a shared structural grammar for interpreting coherence, drift, and environmental coupling across domains. As a result, scientific tools remain siloed, non-interoperable, and difficult to reproduce.

TriadicFrameworks addresses this gap by providing a unified, openly licensed substrate for regime-aware scientific analysis. The framework introduces standardized protocols, structural models, and validation tools that enable consistent interpretation of sensor data, simulation outputs, and cross-disciplinary scientific workflows.

The proposed work advances NASA's open-science mission by delivering transparent, reproducible, and community-accessible infrastructure that supports remote sensing, planetary science, autonomous exploration, and scientific simulation.

---

2. Technical Overview of the TriadicFrameworks Substrate

TriadicFrameworks consists of four interoperable components:

2.1 Dimensional Substrate Regime Scanning Protocol (dsrsp/0.1)

A minimal, substrate-agnostic wire format for regime-aware sensor interpretation.

Key features:

- standardized input envelope for sensor streams
- triadic regime profile (structural, sensory, environmental)
- drift and stability indicators
- optional intelligent-life probability (ILP) module
- RSM and vST alignment layer for downstream engines

dsrsp/0.1 enables consistent interpretation of sensor data across platforms, including remote sensing instruments, ROVs, drones, and simulation environments.

2.2 Resonance Substrate Model (RSM)

A structural grammar for representing coherence, drift, and stability across physical and simulated systems.

RSM provides:

- structural invariants
- coherence signatures
- drift signatures
- stability anchors
- resonance-based structural summaries

RSM serves as the foundation for reproducible structural analysis and cross-domain comparison.

2.3 Validation-Space-Time Engine (vST)

A validation framework for analyzing regime transitions, dimensional continuity, and alignment behavior.

vST includes:

- V1: structural coherence validation
- V2: dimensional continuity validation
- V3: transition and drift validation
- V4: alignment and core-regime validation

The vST engine supports both real-world sensing pipelines and simulation-based training environments.

2.4 Structural Life-Regime Profiles (SLRP)

A standardized set of life-regime exemplars for biological, synthetic, and planetary systems.

SLRP provides:

- structural, sensory, and environmental exemplars
- drift and stability patterns
- cross-domain comparability
- optional ILP integration

SLRP enables consistent interpretation of life-like signatures across sensing modalities.

---

3. Alignment Layer: Integration with RSM and vST Engines

To ensure interoperability, dsrsp/0.1 includes a dedicated alignment layer:

3.1 RSM Structural Envelope

A compact summary for RSM engines:

- coherence_signature
- drift_signature
- stability_signature
- resonance_profile

This enables low-overhead structural classification for simple devices.

3.2 vST Validation Block

A richer feature set for vST engines:

- v1_structural_features
- v2_dimensional_features
- v3_transition_features
- v4_alignment_features

This supports full validation-space-time inference for advanced systems.

3.3 Engine Compatibility Declaration

A simple metadata block:

```text
engine_compatibility:
  rsm: true
  vst: true
  vst_version: "1.x"
```

This ensures downstream systems can automatically determine compatibility.

---

4. Scientific and Technical Objectives

The proposed work will:

1. finalize and document dsrsp/0.1 as an open protocol;
2. produce reference implementations for RSM and vST alignment;
3. develop open-source libraries for regime classification;
4. create example datasets and reproducible workflows;
5. publish integration guides for sensing systems and simulations;
6. support community adoption through documentation and outreach.

These objectives directly support NASA’s goals in open-source science, reproducibility, and cross-disciplinary scientific infrastructure.

---

5. Work Plan and Methodology

5.1 Phase 1 — Protocol Finalization (Months 1–3)

- finalize dsrsp/0.1 specification
- complete RSM and vST alignment layer
- publish schemas and reference examples
- archive all materials with DOIs

5.2 Phase 2 — Reference Implementations (Months 3–6)

- develop open-source libraries for dsrsp/0.1
- implement RSM structural envelope generation
- implement vST validation block generation
- create ILP module reference implementation

5.3 Phase 3 — Integration and Testing (Months 6–9)

- integrate dsrsp/0.1 into simulation environments
- test regime classification workflows
- validate RSM and vST outputs
- produce example datasets and reproducible notebooks

5.4 Phase 4 — Documentation and Community Release (Months 9–12)

- produce developer documentation
- publish integration guides
- release training materials
- host community workshop or virtual tutorial

---

6. Deliverables

The project will deliver:

- dsrsp/0.1 specification and schemas
- RSM and vST alignment layer documentation
- open-source reference implementations
- ILP module
- example datasets and workflows
- developer documentation
- integration guides
- archived DOIs for all components

All deliverables will be openly licensed and publicly accessible.

---

7. Expected Scientific Impact

TriadicFrameworks provides a unified substrate for regime-aware scientific analysis, enabling:

- improved reproducibility across sensing and simulation workflows
- consistent structural interpretation across domains
- enhanced transparency in scientific tooling
- cross-disciplinary interoperability
- community-accessible open-science infrastructure

The framework supports NASA's mission by enabling more robust, transparent, and reproducible scientific analysis across remote sensing, planetary exploration, and autonomous systems.

---

8. Summary

TriadicFrameworks offers a coherent, extensible, and openly licensed substrate for regime-aware scientific analysis. Through dsrsp/0.1, RSM, vST, and SLRP, the framework provides a unified grammar for interpreting structural, sensory, and environmental patterns across scientific domains.

HPOSS support will enable the development, documentation, and community integration of this infrastructure, advancing NASA's open-science mission and supporting future research, exploration, and scientific discovery.